

The Binomial Option Pricing Model

Convergence, Early Exercise, and Market Calibration

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January 31, 2026

Abstract

This report implements a vectorized Binomial Option Pricing Model in Python to evaluate European and American options. The model is validated against the Black-Scholes analytical benchmark, demonstrating asymptotic convergence with discretization error. Empirical calibration to Amazon (AMZN) market data reveals a significant volatility risk premium, with implied volatility ($\approx 42\%$) exceeding historical estimates. Finally, the study confirms that the American Put premium is driven principally by the optimal early exercise boundary in deep in-the-money conditions.

1 Introduction

This project investigates the Binomial and Black-Scholes option pricing models, analyzing their behavior under varying parameters, specifically time steps N and volatility σ . Furthermore, this study extends beyond theoretical derivation to practical application using real-world market data, with a specific focus on the optimal early exercise boundary for American put options. The primary objectives of this project are to:

1. Outline the theoretical framework of both models.
2. Develop a robust Python implementation of the Binomial Tree algorithm for arbitrary steps N and Black-Scholes Pricing Model.
3. Analyze price sensitivity to volatility σ and convergence behavior with increasing N .
4. Validate the models against real-world market data.
5. Investigate the early exercise conditions for American put options.

2 Theoretical Framework

In this project, we introduce two models, the Binomial Option Pricing model and the Black-Scholes model, to price the option.

2.1 The Binomial Option Pricing Model

2.1.1 The Core Concept

The Binomial model, originally proposed by [2], simplifies the complex movement of stock prices by breaking down time into small, discrete steps.

Instead of a continuous curve, we assume that in every short time interval Δt , the stock price S can only move to two possible levels: an up state Su or a down state Sd .

While this "two-state" rule is a simplification, the Law of Large Numbers ensures that as the number of steps increases (making Δt very small), this tree structure converges to the continuous log-normal behavior seen in reality [3].

2.1.2 Risk-Neutral Valuation

Since the value of the option depends entirely on the stock, a risk-free portfolio can be constructed consisting of the underlying asset and a risk-free bond [4]. This no-arbitrage condition yields the risk-neutral probability p :

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (1)$$

where r is the constant risk-free interest rate. To ensure that the model matches the first two moments of the continuous distribution, the up (u) and down (d) factors are parameterized as follows [2]:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}} \quad (2)$$

where σ represents the volatility of the underlying asset.

2.1.3 Backward Induction

Once the tree is built, we calculate the option price from the expiration date to the present. The value of an option at time t is given by the discounted expectation under the risk-neutral measure [2]:

$$V_t = e^{-r\Delta t}[pV_{t+1}^u + (1 - p)V_{t+1}^d] \quad (3)$$

2.1.4 American Options

Unlike European options, American options allow early exercise. Consequently, at each node (t, i) , the option value is defined as the maximum of its intrinsic value (immediate exercise) and its continuation value (holding).

The value of an American Put option at time t is given by [3]:

$$\hat{V}_t = \max(K - S_t, V_t) \quad (4)$$

2.2 Black-Scholes Model

In contrast to the numerical approach of the Binomial Model, Black-Scholes Model provides an analytical solution drawn from stochastic calculus. The model assumes the underlying asset price follows a Geometric Brownian Motion with constant volatility [1].

2.3 Limitations of the Models

2.3.1 Binomial Model

As a discrete-time framework, the Binomial model requires the simulation of multiple potential future price paths. Consequently, it incurs a higher computational cost, particularly when a large number of time steps (N) is required to achieve high precision. However, this numerical approach offers superior flexibility compared to the Black-Scholes model, as its iterative nature can accommodate the early exercise features of American options.

2.3.2 Black-Scholes Model

The primary limitation of the Black-Scholes framework is its reliance on rigid assumptions, most notably that volatility (σ) remains constant throughout the option's life. Since real-market volatility is dynamic, the model requires frequent recalibration to reflect changing market conditions. In contrast, its significant advantage is computational efficiency. As a closed-form analytical solution, it calculates prices with $O(1)$ time complexity, making it instantaneous compared to iterative methods.

3 Python Implementation

The implementation utilizes an Object-Oriented framework to integrate the Binomial and Black-Scholes models. To mitigate the computational inefficiency of iterative loops, the algorithm employs NumPy vectorization for tree generation and backward induction. This approach optimizes performance while accommodating American early exercise conditions (see **Appendix B for implementation**).

3.1 Vectorization Strategy

Instead of iterative loops, we utilized NumPy array broadcasting to calculate the terminal stock prices in a single operation using NumPy broadcasting vectors for the up (u) and down (d) factors.

3.2 Recursive Option Valuation

Following the calculation of the terminal asset prices, the option value is derived via backward induction, proceeding from maturity ($t = N$) to the initial state ($t = 0$). The option value is determined via backward induction, where the holding value is the discounted expectation of the next nodes.

3.3 American Option Extension

To price American options, the algorithm was modified to account for early exercise. At each recursive step, the calculated continuation value is compared against the intrinsic value ($K - S_t$). The option value is updated to be the maximum of these two quantities.

```
1 hold_values = self.df * (qu * values[:-1] + qd * values[1:])
2 current_stock_prices = current_stock_prices[:-1] / u
3 intrinsic_values = self._payoff(current_stock_prices, option_type)
4 values = np.maximum(hold_values, intrinsic_values)
```

This modification ensures that optimal early exercise strategies are captured in the final price.

3.4 Black-Scholes Validation

Finally, the Black-Scholes analytical formula was implemented to serve as a benchmark. Since this closed-form solution is valid only for European options, it provides a standard for verifying the convergence of the Binomial Tree as $N \rightarrow \infty$.

4 Results and Analysis

This section evaluates the Binomial Model against real-market data and the Black-Scholes benchmark (see **Appendix A for details**). The analysis focuses on four key areas:

1. **Empirical Calibration:** Model accuracy and implied volatility derivation for AMZN.

2. **Numerical Convergence:** Discretization error and stability as time steps (N) increase.
3. **Parameter Sensitivity:** The impact of volatility on option premiums (ν).
4. **Early Exercise Boundary:** Analysis of the optimal stopping time for early exercise.

European Call options serve as the primary baseline for analysis, while American Puts are examined specifically to demonstrate early exercise behavior.

4.1 Empirical Test with Market Data

To assess the model’s practical utility, the algorithm was calibrated to real market data for the non-dividend paying stock, AMZN. The initial input parameters, based on historical volatility, are detailed in Table 1.

Parameter	Value
Underlying Asset	AMZN
Date of Observation	24 Jan 2026
Expiration Date	18 June 2026
Time to Maturity (T)	≈ 0.41 years (5 months)
Spot Price (S_0)	\$239.16
Strike Price (K)	\$245.00
Risk-Free Rate (r)	3.768% (6-Month Treasury)
Historical Volatility (σ)	38.18%
Number of Time Steps (N)	50

Table 1: Market Data parameters for AMZN Call Option. Data retrieved from [5].

The pricing results comparing the Binomial Model ($N = 50$), the Black-Scholes benchmark, and the observed market prices are presented in Table 2.

	Historical Volatility		Implied Volatility (Calibrated)	
Metric	Binomial	Black-Scholes	Binomial	Market (Target)
Call Option				
European	22.65	22.56	25.04	24.95
American	22.65	N/A	25.04	24.95
Volatility (σ)	38.18% (Hist)		42.09% (Imp)	
Put Option				
European	9.33	9.34	11.27	11.20
American	9.45	N/A	11.39	> 11.20
Volatility (σ)	38.18% (Hist)		42.03% (Imp)	

Table 2: Comparison of Model Prices: Historical vs. Implied Volatility parameters ($N = 50$).

The results demonstrate strong convergence between the Binomial approximation ($N = 50$) and the Black-Scholes analytical solution. However, a significant discrepancy persists between the calculated model prices and the observed market prices. This deviation is primarily attributed to the input volatility parameter, which was originally derived from historical data rather than current market pricing.

4.1.1 Calibration via Implied Volatility

To reconcile the pricing error, we incorporated the concept of Implied Volatility. Unlike historical volatility, which analyzes past returns, implied volatility represents the market's forward-looking expectation of future variance [3].

We calibrated the model by solving for the volatility (σ_{imp}) required to equate the theoretical Black-Scholes value with the observed market price. Using the Sequential Least Squares Programming (SLSQP) numerical optimization method, we determined the implied volatility was **42.09%** for call option and **42.03%** for put option.

Post-calibration, the Black-Scholes Call price aligns perfectly with the market expectation. The American Put option maintains a premium over the European Put (\$11.39 vs \$11.27), reflecting the value of the early exercise right.

The Volatility Smile "Empirical evidence contradicts the Black-Scholes assumption of constant volatility. In reality, implied volatility varies across different strike prices (K) for a given maturity, often forming a U-shaped curve known as the 'Volatility Smile' [3]. This phenomenon reflects the market's pricing of non-lognormal tail risks.

This phenomenon arises because asset returns do not strictly follow the log-normal distribution assumed by the model; instead, they exhibit "fat tails" and skewness, reflecting a higher probability of extreme market events (or "Black Swans") [3]. To hedge against these tail risks, investors bid up the premiums of deep Out-of-the-Money (OTM) puts, resulting in higher implied volatilities at the extremes compared to At-the-Money (ATM) options.

4.2 Sensitivity of the Option Price to Volatility

This section examines the sensitivity of option premiums to variations in the volatility parameter.

4.2.1 Sensitivity (Vega)

The sensitivity of the option price to volatility is quantified by the Greek letter Vega (ν), defined as the partial derivative of the option value (V) with respect to volatility [3]:

$$\nu = \frac{\partial V}{\partial \sigma} > 0 \quad (5)$$

In the Binomial framework, volatility directly governs the magnitude of the price transitions (u and d). A higher σ expands the lattice of potential future stock prices. The relationship between the Call option price and volatility is illustrated in Figure 1.

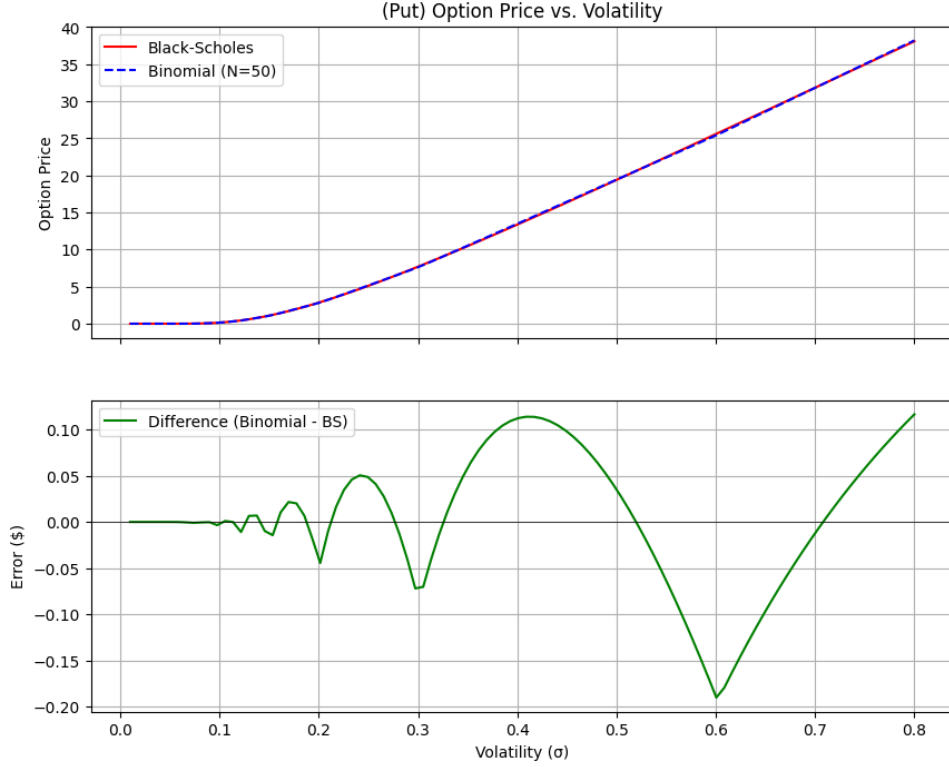


Figure 1: Sensitivity Analysis: Call Option Price vs. Volatility (σ).

The results demonstrate that $\nu > 0$, confirming a positive correlation between volatility and price. The relationship is approximately linear for the observed parameters. The Binomial Model ($N = 50$) exhibits strong agreement with the Black-Scholes benchmark; the discrepancy is negligible ($< 0.5\%$), resulting in visually overlapping curves.

4.2.2 Theoretical Explanation of Vega

The positive relationship between volatility and option value arises from the asymmetric payoff structure (convexity). For an option holder, downside risk is limited to the initial premium paid (limited liability), whereas the upside potential is theoretically unbounded for Call options (and capped only by the strike price for Put options). Higher volatility increases the probability of extreme price movements ("fat tails"). Since the loss is floored at zero while gains are substantial, increased volatility raises the expected value of the option.

4.3 Convergence to Black-Scholes

This section examines the asymptotic convergence of the Binomial Model to the Black-Scholes analytical solution as the number of time steps (N) increases.

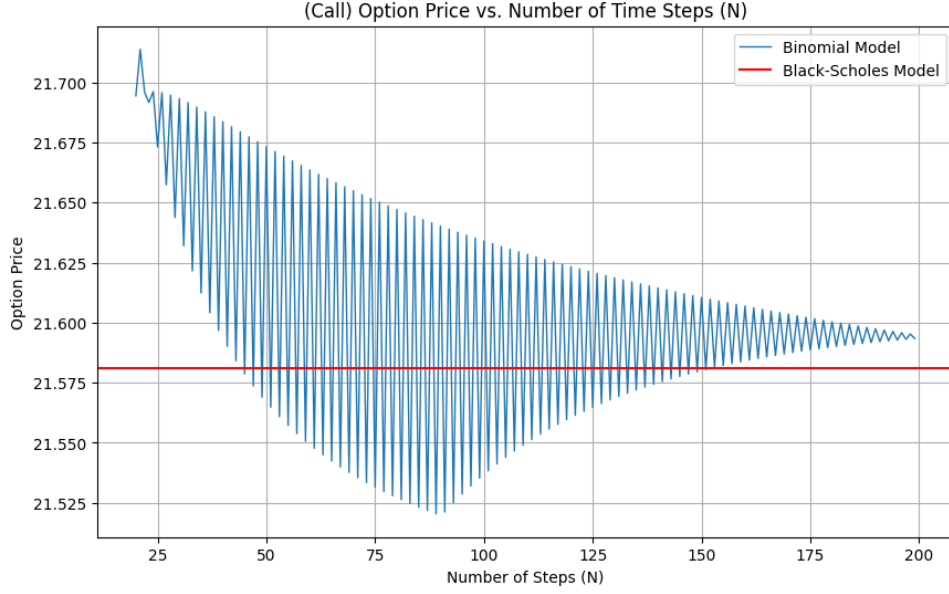


Figure 2: Convergence of Binomial Model Price to Black-Scholes Benchmark ($N = 10$ to 200).

As illustrated in Figure 2, the Binomial price approximates the Black-Scholes value as N increases. However, the convergence is non-monotonic; the trajectory exhibits a distinct oscillatory pattern, particularly at lower values of N .

4.3.1 Analysis of Discretization Error

Unlike the continuous log-normal distribution in Black-Scholes, the Binomial model produces a finite set of terminal stock prices.

The magnitude of the error depends on the alignment of the strike price (K) relative to the terminal nodes. The error is minimized when a node coincides with K and maximized when K falls midway between nodes. Since the node positions shift systematically between odd and even N , the error alternates, creating the observed oscillation.

4.4 Early Exercise Boundary (American Options)

This section analyzes the optimal stopping time for early exercise, visualizing the critical stock price levels that trigger immediate execution.

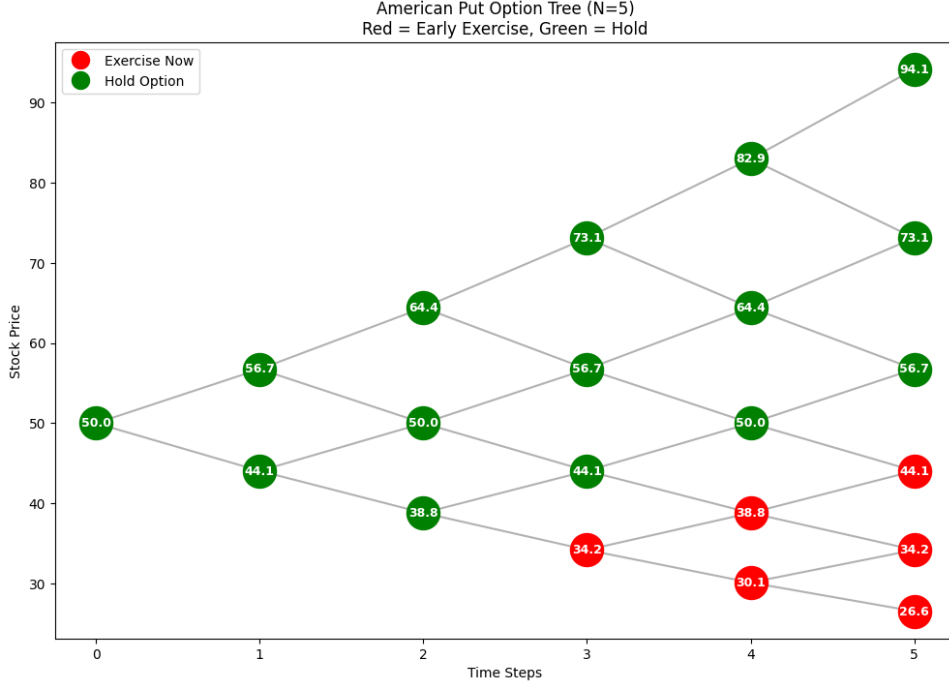


Figure 3: Optimal Early Exercise Boundary for American Put Option. (Visualization generated with the assistance of Google Gemini).

4.4.1 The Optimal Stopping Condition

The decision to exercise early is governed by the comparison between immediate intrinsic value and the expected continuation value. It is optimal to exercise if and only if:

$$K - S_t > e^{-r\Delta t} \mathbb{E}[V_{t+1}] \quad (6)$$

For example, in a deep in-the-money scenario (e.g., $S_t \approx 0$), the remaining downside potential is negligible. In this case, realizing the cash immediately to earn the risk-free rate is superior to holding the option for marginal potential gains. The "boundary" in Figure 3 represents the specific price threshold where these two values are equal.

4.4.2 Asymmetry Between Puts and Calls

For American Call options on non-dividend paying stocks, early exercise is never optimal; thus, the American price equals the European price ($C = c$) [4]. Exercise of a call early requires immediate payment of the strike price K . This results in two losses, Time Value of Money and Insurance Value. In contrast, American Put options often command a premium over their European counterparts ($P \geq p$). Since exercising a Put yields cash (K), deep in-the-money holders may find it optimal to exercise early to receive K and immediately begin earning interest.

5 Conclusion

This project successfully implemented a vectorized Binomial Option Pricing Model, validating its asymptotic convergence to the Black-Scholes benchmark despite inherent discretization error. Empirical calibration to Amazon (AMZN) market data demonstrated that substituting historical volatility with Implied Volatility ($\sigma_{imp} \approx 42\%$) is essential to reconcile model prices with market premiums. Finally, the analysis of the early exercise boundary confirmed that the American Option premium is derived strictly from the optimal stopping condition for deep in-the-money Puts.

References

- [1] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of political economy*, 81(3):637–654, 1973.
- [2] John C. Cox, Stephen A. Ross, and Mark Rubinstein. Option pricing: A simplified approach. *Journal of Financial Economics*, 7:229–263, 1979.
- [3] John C. Hull. *Options, Futures, and Other Derivatives*. Pearson, 8th edition, 2012.
- [4] Sam Marsh. Lecture notes of financial mathematics, 2025. Chapter 4, 5, and 6.
- [5] Yahoo Finance. Amazon.com, inc. (amzn) option chain and historical data, 2026. Accessed: 24 January 2026.

A Google Colab Link

The complete Python code and interactive analysis can be found here:

https://colab.research.google.com/drive/1D-bcokEP3_5GjJWSTjUAssWmqseGSCXe?usp=sharing

B Python Implementation

Core Algorithm: OptionPricer Class

The following class implements the vectorized Binomial Tree and Black-Scholes analytical solution.

```
class OptionPricer:
    def __init__(self, s0, K, r, sigma, T, N):
        self.s0 = s0
        self.K = K
        self.r = r
        self.sigma = sigma
        self.T = T
        self.N = N
        self.dt = T / N
        self.df = np.exp(-r * self.dt)

    def _setup_params(self):
        u = np.exp(self.sigma * np.sqrt(self.dt))
        d = 1 / u
        qu = (np.exp(self.r * self.dt) - d) / (u - d)
        qd = 1 - qu
        return u, d, qu, qd

    def _payoff(self, prices, option_type):
        if option_type == "call":
            return np.maximum(prices - self.K, 0)
        elif option_type == "put":
            return np.maximum(self.K - prices, 0)

    def binomial_price(self, option_style="european", option_type="call"):
        u, d, qu, qd = self._setup_params()

        final_prices = self.s0 * (u ** np.arange(self.N, -1, -1)) * (d ** np.arange(0, self.N))

        values = self._payoff(final_prices, option_type)
        current_stock_prices = final_prices

        for i in range(self.N):
            hold_values = self.df * (qu * values[:-1] + qd * values[1:])

            if option_style == "american":
                current_stock_prices = current_stock_prices[:-1] / u
                intrinsic_values = self._payoff(current_stock_prices, option_type)
                values = np.maximum(hold_values, intrinsic_values)
```

```

        else:
            values = hold_values

    return values[0]

def black_scholes(self, option_type="call", sigma = None):
    vol = sigma if sigma is not None else self.sigma

    d1 = (np.log(self.s0 / self.K) + (self.r + 0.5 * vol ** 2) * self.T) / (vol * np.
    d2 = d1 - vol * np.sqrt(self.T)

    if option_type == "call":
        return self.s0 * norm.cdf(d1) - self.K * np.exp(-self.r * self.T) * norm.cdf(d2)
    elif option_type == "put":
        return self.K * np.exp(-self.r * self.T) * norm.cdf(-d2) - self.s0 * norm.cdf(-d1)

def calculate_implied_volatility(self, market_price, option_type="call"):

    def objective_function(sigma_guess):
        bs_price = self.black_scholes(option_type, sigma=sigma_guess[0])
        error = (bs_price - market_price) ** 2
        return error

    initial_guess = [0.3]
    bounds = [(0.001, 5.0)]

    result = minimize(objective_function, initial_guess, bounds=bounds, method='SLSQP')
    return result.x[0]

```